FEDERAL SPECIFICATION

PAINT, OIL (NONPENETRATING - FLAT, READY-MIXED TINTS AND WHITE)

This specification is approved by the Assistant Commissioner, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers nonpenetrating, flat, ready-mixed oil paint for interior ceilings and walls.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

SS-L-30 - Lath, and Board Products, Gypsum.
TT-S-179 - Sealer, Surface: Pigmented Oil, Plaster and Wallboard.
PPP-P-1892 - Paint, Varnish, Lacquer and Related Materials; Packaging, Packing, and Marking of

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston; New York; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Denver; San Francisco; Los Angeles; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 185 - Test for Coarse Particles in Pigments, Pastes, and Paints.
D 523 - Test for Specular Gloss
D 562 - Test for Consistency of Paints Using the Stormer Viscosimeter.
D 1210 - Test for Fineness of Dispersion of Pigment-Vehicle Systems.

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D 1296 - Test for Odor of Volatile Solvents and Diluents.
D 1729 - Method for Visual Evaluation of Color Differences of Opaque Materials.
D 1849 - Test Method for Package Stability of Paint.
D 2805 - Test Method for Hiding Power of Paints.
D 3335 - Determination of Low Concentrations of Lead in Paint by Atomic Absorption Spectroscopy.
D 3450 - Test Method for Washability Properties of Interior Architectural Coatings.
E 97 - Test for 45-Deg, 0-Deg Directional Reflectance of Opaque Specimens by Filter Photometry.
E 260 - General Gas Chromatography Procedures.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Air Pollution Regulations (SCAQMD)

Rules and Regulations

(Application for copy should be addressed to the South Coast Air Quality Management District, 9150 East Flair Drive, El Monte, CA 91131.)

3. REQUIREMENTS

3.1 Materials. The contractor is given latitude in the selection of ingredients provided the finished paint meets all the requirements specified herein. The solvents used shall be "non-photochemically reactive" as defined in Rule 102 of the South Coast Air Quality Management District regulation in California.

3.2 Quantitative requirements.

3.2.1 The interior nonpenetrating flat paint shall meet the quantitative requirements specified in table I.

TABLE I. Quantitative requirements of nonpenetrating flat paint

Characteristics	Requirement Minimum	Requirement Maximum
Pigment, percent by weight of paint.	56	--
Nonvolatile vehicle, percent by weight of vehicle.	30	--
Water, percent by weight of paint.	--	0.5
Coarse particles and skins (residue retained on U.S. No. 325 sieve), percent by weight of pigment	--	0.5
Fineness of grind.	4	--
Consistency, Krebs-Stormer, K.U.	82	95
Drying time:		
Set-to-touch, hours.	1/2	2
Dry hard, hours.	--	7
85° Specular gloss	--	6
Absorption, inch	--	1/8
Washability; characteristics after test:		
Reflectance, percent of that measured after accelerated aging.	95	--
Gloss, percent of that measured after accelerated aging. .	--	150
Lead, percent by weight of nonvolatile	--	0.06

3.2.2 Opacity. When tested as specified in table IV at a spreading rate of 15.5 m^2/L (630 ft^2/gal), the minimum dry-film contrast ratio in terms of apparent reflectively shall be as specified as in table II.

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TABLE II. Minimum dry-film contrast ratios

Apparent reflectivity (percent)	Contrast ratio
84 and above	0.95
80 - 83	0.96
76 - 79	0.97
70 - 75	0.98
62 - 69	0.99
61 or less	1.00

3.2.3 Resistance to reflectance variation. When tested as in 4.3.7, the general appearance of the panel, including the color of tinted paints, shall be substantially uniform. Any variation in reflectance of the film between the coated sealed and coated unsealed areas shall not exceed 1.0 percent (absolute).

3.3 Qualitative requirements.

3.3.1 Condition in container. The paint shall be thoroughly mixed and ground. It shall not be settled, caked or thickened to such a degree that it cannot be redispersed easily with a paddle or a spatula to a good brushing consistency when tested as specified in table IV.

3.3.2 Color. When tested as specified in 4.3.3., the color shall be a critical match to the specified color of Fed. Std. No. 595.

3.3.3 Storage stability.

3.3.3.1 Partially full container. The paint shall not skin within 48 hours, when tested as in 4.3.4.1.

3.3.3.2 Full container. When tested as specified in 4.3.4.2, the paint shall not liver, cake, skin, or change consistency in excess of that specified in Table I.

3.3.4 Odor. The odor shall not be obnoxious or pungent; and there shall be no odor after 24 hours of drying at room temperature.

3.3.5 Brushing properties. When tested as in table IV, the paint shall brush to a smooth and uniform film, and show no sagging. The dried film shall show no dusting, mottling, or color separation, and shall present a smooth uniform finish, free from seeds.

3.3.6 Spraying properties. The paint, when tested as in table IV at a spreading rate of 12 m^2/L (500 ft^2/gal), shall spray to a smooth and uniform film, and show no running. The dried film shall show no dusting, floating, bubbling, or orange peel.

3.3.7 Roller Coating. When the paint is applied at a spreading rate of 12 m^2/L (500 ft^2/gal) as specified in table IV and allowed to dry in a vertical position, there shall be no evidence of sagging or streaking. The dried film shall show no lapping, localized floating, unevenness of film, or other film defects.

3.3.8 Flexibility. The paint shall show no evidence of cracking when subjected to the flexibility test specified in 4.3.5.

3.3.9 Light Fastness. The paint shall show no darkening or other change in appearance when tested as specified in 4.3.6.

3.4 Marking. The following shall be shown on the reverse side of the container:

NONPENETRATING-FLAT OIL PAINT
(Interior Use)

STIR THOROUGHLY BEFORE USING

This is a washable flat oil paint of the self-priming or self-sealing 'one coat' type, for interior use. It can be applied directly to clean and dry plaster, concrete, brick, wallboard, wallpaper, etc., without a sealing coat. However, if newly patched areas of plaster are adjacent to previously painted surfaces, it is suggested that the patched areas be touched up with the paint, allowed to dry, and then the entire area be given a coat of the paint. The paint is ready-for-use, but if thinning is necessary,

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a small amount (not over 1/2 pint per gallon of paint) of mineral spirits may be added. On some jobs, one coat of this paint may be sufficient. However, on most jobs, two coats are desirable, spreading each coat at the rate of about 500 square feet per gallon, and allowing about 18 hours for drying between coats.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 Lot. For the purpose of sampling, a lot of the paint shall consist of a manufacturer's batch. A batch is defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.2.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with this specification regarding fill, closure, and marking and other requirements not involving tests.

4.2.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of 3.4 and section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.2.4 Sampling of the end item for tests. For the purpose of sampling, the lot shall be expressed in units of gallons. Sample from lots shall be taken in accordance with MIL-STD-105 using inspection level S-2 and an acceptable quality level (AQL) of 2.5.

Tests shall be performed on a sample unit basis. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample unit fails to meet any test requirements specified.

4.3 Test procedures.

4.3.1 Standard test conditions. Unless otherwise specified, all test specimens shall be prepared and tested in a room having a temperature of 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 4 percent.

4.3.2 Tests. The paint shall be tested in accordance with the applicable test methods indicated in table IV, and as specified hereinafter.

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TABLE IV. Test methods

Characteristics	Requirement Reference	Test Methods F.T.M.S.141	Test Methods ASTM	Test Paragraph Reference
Solvent	3.1	--	E 260	--
Condition in container	3.3.1	3011	--	--
Color	3.3.2	--	D 1729	4.3.3
Storage stability	3.3.3.1	3021	--	4.3.4.1
	3.3.3.2	--	D 1849	4.3.4.2
Odor	3.3.4	--	D 1296	--
Brushing properties	3.3.5	4321	--	--
Spraying properties	3.3.6	4331	--	--
Roller Coating	3.3.7	2112	--	--
Flexibility	3.3.8	6221	--	4.3.5
Light fastness	3.3.9	--	--	4.3.6
Resistance to tint variation	3.2.3	--	--	4.3.7
Pigment	Table I	4051	--	--
Nonvolatile vehicle	Table I	4053	--	--
Water content	Table I	4081	--	--
Coarse particles	Table I	--	D 185	--
Fineness of grind	Table I	--	D 1210	--
Consistency	Table I	--	D 562	--
Drying time	Table I	4061	--	--
Absorption	Table I	4421	--	--
85° Specular gloss	Table I	--	D 523	4.3.10
Washability	Table I	--	D 3450	4.3.9
Dry opacity	Table II	--	D 2805	--
Lead content	Table I	--	D 3335	4.3.8

4.3.3 Color. Apply the paint on a clean smooth plate glass panel with a 76 um (0.003-inch) film applicator and allow to dry for 24 hours. Recoat if necessary to obtain complete hiding. Determine compliance with 3.3.2 by ASTM D 1729.

4.3.4 Storage stability.

4.3.4.1 Partially full container. Fill a 250 ml (8-ounce) container with 188 ml (6-ounce) of the paint, seal, and test in accordance with method 3021 of Fed. Test Method Std. No. 141. After 48 hours examine for compliance with 3.3.3.1.

4.3.4.2 Full container. Test a sealed pint container of paint at 52±1°C (125±2°F) for 3 weeks in accordance with ASTM D 1849. Evaluate for compliance with 3.3.3.2.

4.3.5 Flexibility. Apply a film with a 102 um (0.004-inch) gap clearance film applicator on an electrolytic tin panel prepared in accordance with method 2012 of Fed. Test Method No. 141. Air dry for 18 hours and bake for 5 hours at 105 + 1°C (221 + 2°F.). Cool at room temperature for 1/2 hour, bend over a 6.35 mm (1/4-inch) mandrel, as specified in method 6221 of Fed. Test Method Std. No. 141. Evaluate for compliance with 3.3.8.

4.3.6 Light fastness. Apply a sufficient number of coats of the paint to a ground glass plate to hide the surface completely. Allow each coat to dry for 24 hours. Cover half of this painted surface with black paper sandwiched between two sheets of white paper, and expose the coated surface to G.E. sun lamp (110-125 v.) or equal at 12 inches horizontal position for 6 hours. Remove the paper and examine the surface. Observe for compliance with 3.3.9.

4.3.7 Resistance to reflectance variation. Apply the paint to a penetration chart to a wet film thickness of 76 + 2 um (0.003 + 0.0001 inch) and width of 89 mm (3 1/2 inches). Allow the material to dry in a horizontal position for 24 hours at room temperature. After the drying period, obtain 45° directional reflectance measurements in accordance with ASTM E 97 on the sealed and unsealed portions of the penetration chart, and report the difference. Evaluate for compliance with 3.2.3.

4.3.8 Lead Content. Determine lead content in accordance with ASTM D 3335 or as specified below. In case of dispute, the following procedure shall be used:

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4.3.8.1 Procedure. Using a 152 um (0.006-inch) gap film applicator, duplicate drawdowns of the well-mixed paint shall be made on the sealed portion of standard paint penetration charts, dried for 24 hours, and cut to fit the sample holder of a fluorescence X-ray spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight of the total non-volatile. Settings for a wavelength dispersive spectrometer shall be:

Element	Analytical Line	Angle	Crystal	Detection	Collimator	X-ray tube (Mo)
Pb	L	33.93	LiF(200)	Flow S.C.	Fine	60kV 45mA
Pb (background I)		33.00	LiF(200)	Flow S.C.	Fine	60kV 45mA
Pb (background II)		33.50	LiF(200)	Flow S.C.	Fine	60kV 45mA
Mo	K	20.33	LiF(200)	Flow S.C.	Fine	60kV 45mA

Pulse height selection shall be used in all measurements and counting time shall be 100 seconds. Place the sample disc in the wavelength dispersive unit. Measure the count rates of lead, lead background, and Molybdenum Compton scattered background from the X-ray tube.

4.3.8.2 Calculation.

$$R = \frac{2I_{Pb} - I_{Pb}(I) - I_{Pb}(II)}{2I_{Mo}}$$

where I equals gross intensity. These results shall be compared with those obtained by a 0.06 percent lead standard made up from the same type of paint sample and evaluated for compliance with the requirement in Table I.

4.3.9 Washability. The washability shall be tested in accordance with ASTM D 3450 using abrasive medium and evaluated for conformance with Table II.

4.3.10 85° specular gloss. Prepare a film by drawing down with a film applicator a wet film thickness of 76 + 2 um (0.003 + 0.0001 inch) on a plane opaque white glass panel as prescribed in 2.1.5 of method 2021 of Federal Test Methods Standard 141. Air dry for 24 hours at standard condition. Determine 85° specular gloss in accordance with ASTM D 523 for compliance with table I.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with PPP-P-1892. The levels of packaging shall be A or C and packing shall be A, B, or C, as specified (see 6.2). The paint shall be furnished in a 1-quart can, 1-gallon can, 5-gallon pail, or 55-gallon drum (see 6.2).

6. NOTES

6.1 Intended use. Flat wall paint covered by this specification is intended for use, as received on interior walls and ceilings. It is designed for direct application over porous surfaces such as wallboard, wallpaper, plaster, or other absorptive materials, to give a uniform finish without the use of a special primer or sealer.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1).
(c) Color requirement (see 3.3.2).
(d) Level of packaging and packing required (see 5.1).

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6.3 Paint covered by this specification has good washability for a flat paint, high dry hiding, and good sealing properties. While this is a performance type of specification, it is contemplated the manufacturer will use a very long-oil varnish vehicle. It should consist mainly of properly processed drying oils together with a small amount of volatile thinners and driers. The oil length of the vehicle should be at least 100 gallons. The usual opaque white pigments such as titanium dioxide, titanium-base pigments, or zinc-sulfide-base pigments, together with suitable extending pigments should be used. Tinting colors should be resistant to alkali, since the paint may be used directly on unprimed plaster.

6.4 Flat wall paint should be purchased by volume (3.785 liters or 231 cubic inches at 15°C (60°F) equals 1 U.S. gallon)

CIVIL AGENCY COORDINATING ACTIVITIES:

USDA - U.S. Forest Service
COMM - NBS
VA - OSS
GSA - FSS

PREPARING ACTIVITY:

GSA - FSS

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